Geometric Series formulae:

⇒ Infinite length geo series:

$$P = \sum_{n=0}^{\infty} a^n \Rightarrow P = \frac{1}{1-a}, |a|<1$$

⇒ finite length geo series:

$$Q = \sum_{n=0}^{L} a^n \Rightarrow Q = \frac{1-a^{L+1}}{1-a}$$

$$Q = \sum_{n=L_1}^{L_2} a^n \Rightarrow Q = \frac{a^{L_1} - a^{L_2+1}}{1-a}$$

Normalised Energy:

Cont:

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt$$

Dist:

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Normalised Power:

Cont:

Periodic: $P_x = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x^2(t) \cdot dt$

Non-Periodic: $P_x = \lim_{T\to\infty} \left[\frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t) \cdot dt \right]$

DT:

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n]$$

$$P_x = \lim_{M\to\infty} \left[\frac{1}{2M+1} \sum_{n=-M}^{M} x^2[n] \right]$$

Even Symmetry:

$$x[-n] = x[n] \Rightarrow x_e = \frac{x[n] + x[-n]}{2}$$

Odd Symmetry:

$$x[-n] = x[n] \Rightarrow x_o = \frac{x[n] - x[-n]}{2}$$

Periodicity:

DT:

Ex $\cos(2\pi F_0 N)$

$$F_0 = \frac{2\pi F_0}{2\pi}$$

$$N = \frac{k}{F_0}$$

$\boxed{k = int}$

Causality in DT:

$h[n] = 0$ for all $n<0$

Stability in DT:

$$\sum_{k=-\infty}^{\infty} [h[k]] < \infty$$

Superposition:

$$y[n] = Sys\{\alpha_1 x_1[n] + \alpha_2 x_2[n]\}$$

$$= \alpha_1 y_1[n] + \alpha_2 y_2[n]$$ ✓ (linear)

✗ (non-linear)